ControlPlane.ai — Admission-Control Layer for AI That Acts

Accenture Innovation Challenge 2026 · Round 2 · Team ControlPlane · PS #1

Public GitHub: https://github.com/Srujan0798/controlplane-ai · branch main · tag v0.2.0-round2

1. Portal uploads (five fields)

These are the only artifacts the submission portal needs. This PDF is field 3.

#	Portal field	Artifact
1	Public GitHub link	https://github.com/Srujan0798/controlplane-ai
2	Prototype video	submission/ControlPlane_Round2_Prototype.mp4
3	README document (PDF)	submission/ControlPlane_Round2_README.pdf
4	Business proposal (PDF)	submission/ControlPlane_Round2_Proposal.pdf
5	Pitch deck (PPTX)	submission/ControlPlane_Round2_Pitch.pptx

2. What this is

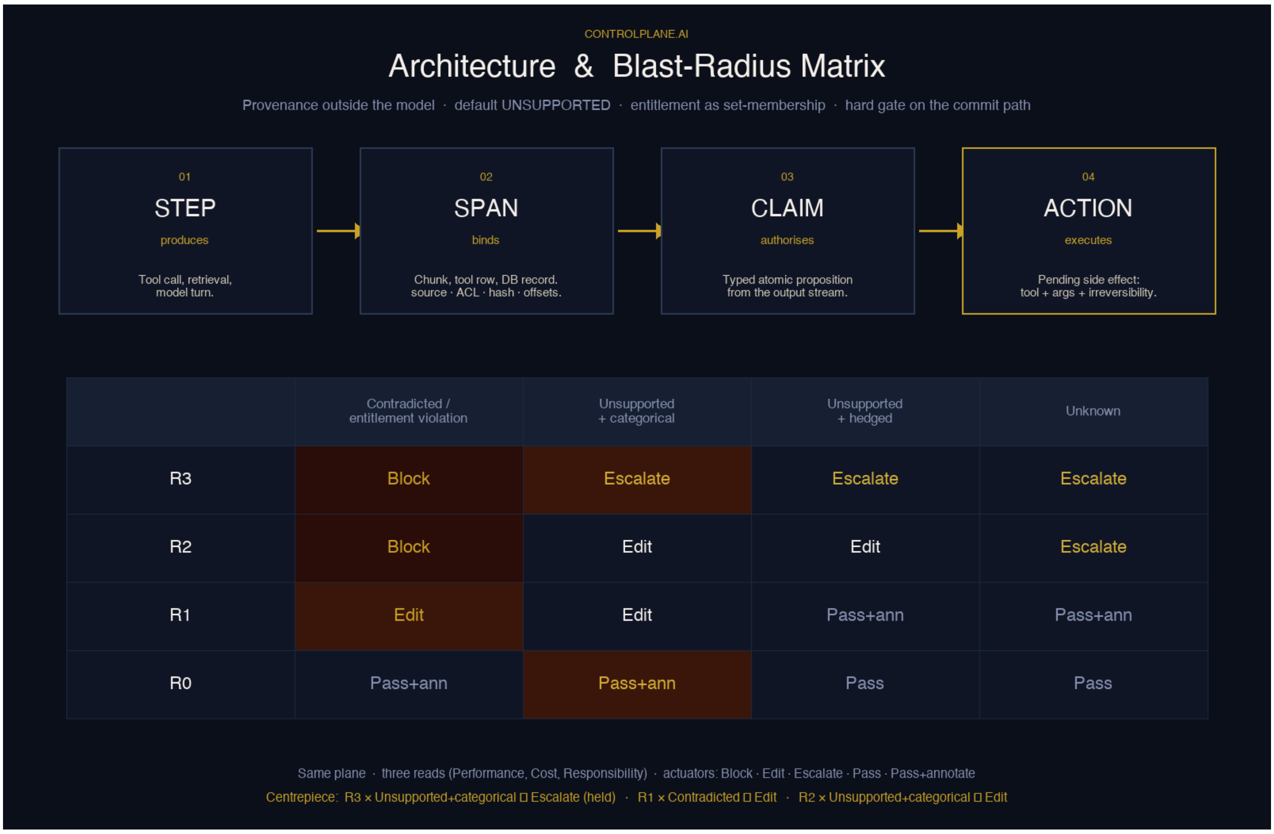
Every AI response that asks permission to **act** — issue a refund, send an email, publish a record — is a set of **claims**. ControlPlane captures provenance **outside** the model (STEP → SPAN → CLAIM → ACTION) and admits or holds each action through a frozen interlock matrix. Unproven claims cannot authorize irreversible actions. Lane 1 is deterministic; no LLM sits on the critical path.

This repository is the working prototype. The numbers in the evidence section are produced by `make eval` and `make bench` and are regenerated on every run — they are measured, not asserted.

3. The mechanism

Run any response text through extract → bind → entitle → interlock → shadow:

• **extract** — Lane-1 segmenter produces typed claims (numeric / structural / temporal / textual / derived) with hedging and role-in-action, no LLM.
• **bind** — each claim is matched to provenance spans: BM25 + lexical gate (textual), Indian/Western numeric normalization (numeric), symbol-table lookup (structural), recompute (derived). Middle band → UNKNOWN, never a default pass.
• **entitle** — span ACL is checked against principal clearance; an unbound PII entity is a leak (Rule A) → Block at R2/R3.
• **interlock** — a frozen 16-cell matrix maps (blast-tier × verdict-column) to one of five actuators. The matrix is never redrawn.
• **shadow** — every decision is dual-emitted; the published FNR carries a Wilson interval.



Architecture (STEP → SPAN → CLAIM → ACTION) and the frozen blast-radius matrix. Same response, two actuators: show_text → Edit (R1) · issue_refund → Escalate, **held** with the evidence packet — never 'blocked'.

4. Run it in 60 seconds (clean clone)

```
git clone https://github.com/Srujan0798/controlplane-ai && cd controlplane-ai
python3 -m venv .venv && source .venv/bin/activate
pip install -e ".[dev]"
bash scripts/preflight-lite.sh          # must PASS
make run                                # http://127.0.0.1:8787
# refund:  http://127.0.0.1:8787/?scenario=refund&mode=enforce&autorun=1
# expected: show_text → Edit · issue_refund → Escalate (held)
# gate:     http://127.0.0.1:8787/gate   (arbitrary text, no fixtures)
make eval                               # FNR with Wilson CI
make verify                             # tests + content laws + number freeze
```

Refund language is **held and escalated with the evidence packet** — never 'blocked'.

5. What to look at

• **/gate** — paste a response; watch claims, binding method + rationale, symbol table, matrix cell, actuator, evidence packet, dead compute, per-stage latency.
• **refund autorun** — show_text → Edit · issue_refund → Escalate (held).
• **examples/refund_trace_demo.py** — same dual-action with fixtures off; amount via numeric.
• **evals/cases/** — labelled cases incl. hard negatives; this is the corpus a judge can attack.
• **controlplane/interlock.py** MATRIX — the sixteen transcribed cells, sole decider.

6. Capability ledger

Every mechanism in the architecture, marked honestly: **implemented and tested** / **implemented, prototype-grade** / **designed, not built**. This preempts 'is any of this real?' by answering first.

Mechanism	Status	Where
Claim extraction (Lane 1, no LLM)	implemented and tested	controlplane/extract.py
Numeric normalization (IN/US grouping)	implemented and tested	controlplane/numeric.py
BM25 + lexical entailment gate	implemented and tested	controlplane/bm25.py
Symbol table / structural lookup	implemented and tested	controlplane/symbols.py
Binder v2 route dispatch	implemented and tested	controlplane/binder.py
PII Rule A (leak → Block)	implemented and tested	controlplane/pii.py
Entitlement / ACL audit	implemented and tested	controlplane/entitlement.py
Frozen 16-cell interlock matrix	implemented and tested	controlplane/interlock.py
Lane deadlines (async gather)	implemented and tested	controlplane/lanes.py
Hash-chained evidence ledger	implemented and tested	controlplane/ledger.py
Override HTTP → chained ledger	implemented and tested	controlplane/server/app.py
Analyze arbitrary text	implemented and tested	POST /v1/controlplane/analyze
OpenAI-compatible FastAPI gate	implemented and tested	controlplane/server/app.py
Dual-action refund (Edit + Escalate held)	implemented and tested	examples/refund_trace_demo.py
Eval harness + Wilson CIs	implemented and tested	evals/run.py
Load bench n=10000	implemented and tested	scripts/load_bench.py
published_fnr from last eval	implemented and tested	GET /v1/controlplane/metrics
Shadow-replay threshold gate	implemented and tested	controlplane/feedback.py
Canary auto-rollback	implemented and tested	controlplane/feedback.py
Jurisdiction packs (EU/IN/GDPR)	implemented and tested	policies/*.yaml
Multi-turn claim inheritance	implemented and tested	controlplane/session.py
Economist / dead-compute walk	implemented and tested	controlplane/economist.py
Bias counterfactual (Lane 3)	implemented, prototype-grade	controlplane/bias.py
GPU / trained NLI adapter	designed, not built	Lane-2 slot (stub)
Production FNR on live traffic	designed, not built	shadow earn-out
Production IAM repair	designed, not built	out of Round 2 scope

7. Evidence (measured on this run)

Eval corpus: **168** labelled cases (426 bindings), **self-authored**. Ungrounded FNR **1.1%** (95% CI 0.2%–5.8%; n=94 ungrounded). We HOLD **98.9%** of ungrounded responses. Passable-action false-hold (FPR) **0.0%** (95% CI 0.0%–15.5%). Fail-closed caution on hard-negatives **64.2%**. Abstention (UNKNOWN rate) **0.5%** of bindings.

Published miss: `struct-miss-000` (see evals/last_run.json).

Per-route binding distribution:

Route	n	supported	unknown	contradicted	unsupported	abstain
numeric	119	105	0	14	0	0.0%
structural	211	178	0	0	33	0.0%
textual	71	50	2	0	19	2.8%

derived	12	2	0	10	0	0.0%
temporal	11	11	0	0	0	0.0%
none	2	0	0	2	0	0.0%

Load bench: n=10000 at concurrency sweep [1, 4, 16, 64]. Per-request compute p50=0.4217 ms, p95=0.4806 ms, p99=0.6039 ms, mean=0.4309 ms, max=8.65 ms. Best throughput 2292.0 rps. (In-process compute, not API latency; 40ms is a Lane-1 target, never a measured p95.)

Refund language in this system is **held and escalated with the evidence packet** — never 'blocked'. Irreversible actions are not released without provenance.

7.1 Honest reading of these numbers

Measured on a **168-case self-authored corpus**. Ungrounded FNR **1.1%** (95% CI 0.2%–5.8%), passable FPR **0.0%** (95% upper bound 15.5%). Hard-negative hold rate is **64%** (64.2%, 95% Wilson CI 50.7%–75.7%; 53/168 = 31.5% hard negatives) — we over-flag, we measured it, and it is our named next milestone. No production traffic; production FNR is unknown. The corpus ships in the repo so you can attack it.

7.2 What we refuse to claim

This list is about **us**, not about anyone else. **We do not claim:** elimination of hallucinations · zero integration cost · zero added latency · one accuracy number · production-scale proof.

8. Repo map

controlplane/ — pipeline (extract, binder, symbols, bm25, numeric, pii, entitlement, interlock, lanes, feedback, bias, shadow, server)
evals/ — corpus (cases/), harness (run.py), Wilson CIs, last_run.json
policies/ — jurisdiction packs (eu / in / gdpr / default)
scripts/ — bench, proposal/readme PDF builders, preflight, verify
examples/ — refund_trace_demo, knowledge_flip_demo, multi_usecase_demo
submission/ — five portal files: mp4 · README.pdf · Proposal.pdf · Pitch.pptx
Public GitHub — https://github.com/Srujan0798/controlplane-ai

Content laws (ARCHITECTURE §10) are enforced in CI: pytest tests/test_content_laws.py.

9. Buyers (target users)

Five personas this product is for, in the order the brief asks:

• **Customer-support agents** — R1 — Edit on the user-visible reply. Strip unentitled spans (PII, internal notes) before the customer sees them. The incident output-only checkers cannot see.

• **Internal copilots in regulated workflows** — R2 — Edit on reversible writes and external sends. Autonomy downgrade: draft becomes suggestion, never send.

• **Decision-support & acting systems** — R3 — Escalate, held with the evidence packet, on payments, deletions, publications, regulated advice. Fail closed.

• **Compliance & risk teams** — Hash-chained evidence ledger, per-route FNR with Wilson intervals, audit packets downloadable per request. Empty schema until measured.

• **Platform & security teams** — OpenAI-compatible proxy + SDK hook at context assembly. Jurisdiction packs (EU / IN / GDPR) as DAG content. No model access, no weights, no fine-tune.

10. Roadmap (phased, with exit criteria)

Enforcement is earned per route. Nothing goes live until shadow has produced its own counterfactual.

Phase	Window	Exit criterion
Phase 0 — Shadow	weeks 0–6	Proxy + SDK hook at context assembly; dual-emit. Deliverable: would-have-held N, of which M we
Phase 1 — Enforce R3	weeks 6–12	Irreversible actions (payments, deletion, publication, regulated advice). Fail closed or escalate.
Phase 2 — Enforce R2	weeks 12–20	Reversible writes and external sends. Autonomy downgrade.
Phase 3 — FNR + loops	from week 16	Per-route FNR with confidence intervals; override capture; jurisdiction packs as DAG content; cou

11. Risks (with mitigations)

Six risks we name out loud, because a control plane that hides its failure modes is indistinguishable from false assurance.

Risk	Mitigation
Parametric / no-evidence answer	Declared ungrounded by construction; semantic-entropy probe runs async on the calibration lane, never on the
Multi-hop / derived claim	Anything arithmetic or aggregative is recomputed from spans. Anything neither recomputable nor entailed → U
Over-permissioned source index	We faithfully enforce whatever ACL the source carries — and we log every entitlement decision against a name
Prompt injection	Binding is computed by us, not asserted by the model — the model has no channel to declare a binding. The s
Plane failure	Fail stance is declared per blast-radius tier, not globally. R0/R1 fail open with annotation; R2/R3 fail closed or e
Production FNR unknown	The honest answer. The only number on the slide is measured on a 168-case self-authored corpus. Production

12. Team

Choda Srujan Sai — architecture, mechanism, demo build, latency and eval. **Dhrithika** — narrative, pitch, hostile-Q&A;, presentation. **IIT Gandhinagar · Team ControlPlane · Accenture Innovation Challenge 2026 · PS #1**

Two-person team. The repo is the proof: 36 modules, 50+ tests, frozen matrix, deterministic rebuild of every portal artifact from measured JSON.

13. Glossary

Term	Definition
STEP	tool call, retrieval, or model turn that produces spans.
SPAN	chunk / tool row / DB record captured at context assembly — source_id · ACL · hash · offsets. Frozen before generat
CLAIM	typed atomic proposition from the output stream (numeric / structural / temporal / textual / derived).
ACTION	pending side effect — tool + args + irreversibility tier.
R-tier (blast radius)	R0 read-only · R1 user-visible text · R2 reversible writes · R3 irreversible.
Matrix	frozen 16-cell f(R, S) → actuator. Transcribed from the canon, never redrawn.
Set-membership	span.acl ⊆ principal.clearance — pure rule, zero LLM.
Default UNSUPPORTED	a claim must earn SUPPORTED. Absence of evidence is not conflicting evidence.
Held ≠ blocked	the refund is held and escalated with the evidence packet, never 'blocked'. Spoken word and grid agree.
Wilson CI	95% confidence interval on a proportion — the only honest way to publish rates from a finite corpus.